Next: relate Rm to $\tilde{Rm}$ for $M \subset \tilde{M}$

Weingarten Formula: For $X, Y \in \mathcal{T}(M)$, $N \in \mathcal{N}(M)$

$$\langle \tilde{\nabla}_X N, Y \rangle = - \langle N, \mathrm{II}(X,Y) \rangle$$

Proof: On M, $\langle N, Y \rangle = 0$

$$\Rightarrow 0 = X \langle N, Y \rangle$$

$$= \langle \tilde{\nabla}_X N, Y \rangle + \langle N, \tilde{\nabla}_X Y \rangle$$

$$= \langle \tilde{\nabla}_X N, Y \rangle + \langle N, \mathrm{II}(X,Y) \rangle$$

Thm (Gauss equation)

For $X, Y, Z, W \in \mathcal{T}(M)$,

$$Rm(X,Y,Z,W) = \tilde{Rm}(X,Y,Z,W) + \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z) \rangle - \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W) \rangle$$

Proof: On M,

$$\tilde{\nabla}_X \tilde{\nabla}_Y Z = \tilde{\nabla}_X (\nabla_Y Z + \mathrm{II}(Y,Z)) \quad \text{Gauss formula}$$

$$\Rightarrow \langle \tilde{\nabla}_X \tilde{\nabla}_Y Z, W \rangle = \langle \nabla_X \nabla_Y Z, W \rangle - \langle \mathrm{II}(Y,Z), \mathrm{II}(X,W) \rangle \quad \text{Weingarten}$$

sim. for $X \leftrightarrow Y$

Also $\langle \tilde{\nabla}_{[X,Y]} Z, W \rangle = \langle \nabla_{[X,Y]} Z, Y \rangle$ on M

Special case: surface $\Sigma \subset \mathbb{R}^3$

Σ is orientable if $\exists$ unit normal vector field
$\Rightarrow N\Sigma \cong \Sigma \times \mathbb{R}$ (trivial)

With N fixed, we can write $\underline{II}$ in a scalar form

$h(X,Y) = \langle \underline{II}(X,Y), N \rangle$ sym. $\binom{2}{0}$-tensor

or $\underline{II}(X,Y) = h(X,Y)N$

By the Weingarten formula,

$h(X,Y) = -\langle \bar{\nabla}_X N, Y \rangle$ for vector fields X, Y on Σ

In coords., $\bar{\nabla}_{\partial_j} N = N_j$ (as a vector in $\mathbb{R}^3$)

$\Rightarrow \bar{\nabla}_X N = dN(X)$

Weingarten becomes $h(X,Y) = \langle -dN(X), Y \rangle$

The Gauss curvature was $K = \det dN$

Given an orthonormal basis for T_pM $\{E_1, E_2\}$

$K = \det\left[h(E_i, E_j)\right]$

For $\dim = 2$, Rm has only one indep. function and the Gauss eq. gives

$$Rm(E_1,E_2,E_2,E_1) = \underbrace{\bar{Rm}(\dots)}_{0 \text{ for } \mathbb{R}^3} + \langle \underline{II}(E_1,E_1), \underline{II}(E_2,E_2) \rangle - \|\underline{II}(E_1,E_2)\|^2$$

$$= K(p)$$

We can reconstruct $Rm(\cdot,\cdot,\cdot,\cdot)$ using symmetries

The resulting formula applies to any surface, since Rm is a local isometry invariant,

Thm For a Riemannian surface Σ, $\exists$ a function $K \in C^\infty(\Sigma)$ such that

$$Rm(X,Y,Z,W) = K \cdot [\langle X,W\rangle\langle Y,Z\rangle - \langle X,Z\rangle\langle Y,W\rangle]$$

(or $R_{ijkl} = K(g_{il}g_{jk} - g_{ik}g_{jl})$)

Cor For surfaces: $Ric(X,Y) = K\langle X,Y\rangle$ $\quad (R_{ij} = Kg_{ij})$

and $S = 2K$.

Sectional curvature

Given $p \in M$, a plane $\Pi \subset T_pM$ defines a 2-dim. submfld near p by

$$\Pi \longmapsto S_\Pi = \exp_p(\Pi)$$

def.: The <u>sectional curvature</u> is the function

$$\Pi \longmapsto K(\Pi) = \text{Gauss curvature of } \Sigma \text{ at } p$$

Let Rm = curvature of M, $\overline{Rm}$ = curvature of Σ

If X, Y span Π then by the Gauss curvature formula,

$$K(\Pi) = \underbrace{\frac{\overline{Rm}(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2}}_{=: K(X,Y)}$$

In fact we can use Rm in place of $\overline{Rm}$ here

<u>Prop.</u> For X, Y a basis for the plane $\Pi \subset T_pM$,

$$K(X,Y) = \frac{Rm(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2}$$

Proof: Use g for M, $\bar{g}$ for $\Sigma = \exp_p(\Pi)$

Claim: <u>Π</u> vanishes on $T_p\Sigma$

For $V \in \Pi$, let $\gamma(t) = \exp_p(tV)$

γ geodesic in $M \Rightarrow D_t\dot\gamma = 0$

By the Gauss formula

$$D_t\dot\gamma = \overline{D}_t\dot\gamma + \underline{\Pi}(\dot\gamma, \dot\gamma)$$

Since $\bar{D}_t \dot{\gamma} \in TM$ and II is normal

$D_t \dot{\gamma} = 0 \Longrightarrow \bar{D}_t \dot{\gamma} = 0$ and $\mathrm{II}(\dot{\gamma}, \dot{\gamma}) = 0$

evaluating at $t=0$ gives $\mathrm{II}(v, v) = 0$

$\Rightarrow \mathrm{II} = 0$ on $T_p M$ since II is symmetric

By the Gauss equation

$$Rm(\cdot,\cdot)\cdot = \overline{Rm}(\cdot,\cdot)\cdot + (\text{terms involving } \mathrm{II})$$

$\therefore Rm(\cdot,\cdot)\cdot = \overline{Rm}(\cdot,\cdot)\cdot$ on $T_p\Sigma$.

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Note: the argument only shows $\mathrm{II} = 0$ at p, not at other pts. of Σ.